

Preconditioning and Numerical Stability in Neural Network Training for Parametric PDEs

Chenguang Duan
RWTH Aachen

Joint work with **Markus Bachmayr** (RWTH Aachen),
Wolfgang Dahmen (USC & RWTH Aachen), and **Mathias Oster** (RWTH Aachen)

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Outline

- ➊ Neural Network Methods for Parametric PDEs
- ➋ Frame Representation and Condition number
- ➌ Numerically Stable Preconditioning
- ➍ Numerical Experiments
- ➎ Conclusions

Parameter-dependent PDEs in the residual form

$$\mathcal{R}(u(y); y) = 0.$$

- ▶ Parameter $y \in Y$.
- ▶ Spatio-temporal variables $x \in \Omega \subset \mathbb{R}^d$.
- ▶ Parameter-to-solution map $y \mapsto u(y) = u(\cdot, y)$.

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Challenge: Linear model reductions face significant challenges when the Kolmogorov n -widths of the solution manifold decay slowly. $\rightsquigarrow$ **nonlinear reduced models, e.g., neural networks.**

Pure neural network approximation

$$u(x, y) \approx u^{\text{NN}}(x, y; \theta), \quad \forall (x, y) \in \Omega \times Y.$$

- ▶ difficult to satisfy the boundary condition
- ▶ poor theoretical understanding

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Hybrid approximation

$$u(x, y) \approx \sum_{i \in \mathcal{I}} u_i^{\text{NN}}(y; \theta) \varphi_i(x), \quad \forall (x, y) \in \Omega \times Y.$$

- ▶ fixed set of functions $\Phi := \{\varphi_i\}_{i \in \mathcal{I}}$ on the physical domain Ω with $|\mathcal{I}| < \infty$;
- ▶ neural network: $y \mapsto \mathbf{u}^{\text{NN}}(y; \theta) := (u_i^{\text{NN}}(y; \theta))_{i \in \mathcal{I}}$.

Weak Formulation and Loss Functions

Let U the trial space and V be the test space. Consider a linear parametric PDE

$$\mathcal{R}(u(y); y) := \mathcal{B}_y u(y) - f = 0, \quad y \in Y,$$

where $\mathcal{B}_y : U \rightarrow V'$ is a parameter-dependent bounded linear operator, and $f \in V'$.

Weak formulation

For any $y \in Y$, find $u(y) \in U$ such that

$$\mathcal{R}(u(y); y)(v) = (\mathcal{B}_y u(y))(v) - f(v) = 0, \quad \forall v \in V.$$

Weak Formulation and Loss Functions

Well-posedness: Banach–Nečas–Babuška (BNB) conditions

There exist constants $0 < c_b \leq C_b < \infty$, uniform for $y \in Y$, such that

$$\begin{aligned} \sup_{w \in U \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{(\mathcal{B}_y w)(v)}{\|w\|_U \|v\|_V} &\leq C_b, & \inf_{w \in U \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{(\mathcal{B}_y w)(v)}{\|w\|_U \|v\|_V} &\geq c_b, \\ \sup_{w \in U} (\mathcal{B}_y w)(v) &> 0, & \forall v \in V \setminus \{0\}. \end{aligned}$$

$$c_b \|w - u(y)\|_U \leq \|\mathcal{R}(w; y)\|_{V'} \leq C_b \|w - u(y)\|_U, \quad \forall w \in U, y \in Y.$$

Weak Formulation and Loss Functions

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$$c_b \|w - u(y)\|_U \leq \|\mathcal{R}(w; y)\|_{V'} \leq C_b \|w - u(y)\|_U, \quad \forall w \in U, y \in Y.$$

► Residual-based loss

$$\min_{w \in U} \|\mathcal{R}(w; y)\|_{V'}^2 = \|\mathcal{B}_y(w) - f\|_{V'}^2 \approx \|w - u(y)\|_U^2.$$

► Data-driven loss

$$\min_{w \in U} \|w - u(y)\|_U^2.$$

Condition Number and Optimization

Finite-dimensional trial space $U_\Phi := \text{span } \Phi$.

Data-driven loss posed in U_Φ

$$\min_{w \in U_\Phi} \|w - u(y)\|_U^2 = \min_{\mathbf{w} \in \mathbb{R}^{|\mathcal{I}|}} \left\{ \langle \mathbf{A}\mathbf{w}, \mathbf{w} \rangle_{\ell_2} - 2\langle \mathbf{f}, \mathbf{w} \rangle_{\ell_2} \right\}, \quad \mathbf{A}_{ij} = \langle \varphi_i, \varphi_j \rangle_U.$$

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$$\min_{w \in U_\Phi} \|\mathcal{B}_y w - f\|_{V'}^2 = \min_{\mathbf{w} \in \mathbb{R}^{|\mathcal{I}|}} \left\{ \langle \mathbf{A}_y \mathbf{w}, \mathbf{w} \rangle_{\ell_2} - 2\langle \mathbf{f}_y, \mathbf{w} \rangle_{\ell_2} \right\}, \quad (\mathbf{A}_y)_{ij} = \langle \mathcal{B}_y \varphi_i, \mathcal{B}_y \varphi_j \rangle_{V'}.$$

BNB conditions imply: $\text{cond}(\mathbf{A}_y) \leq \frac{C_b^2}{c_b^2} \text{cond}(\mathbf{A})$.

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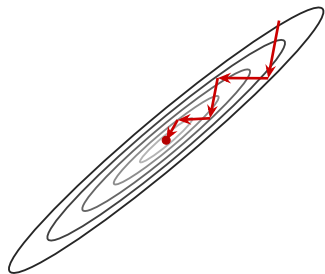
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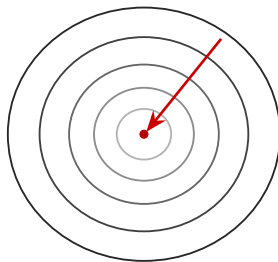
BNB conditions imply: $\text{cond}(\mathbf{A}_y) \leq \frac{C_b^2}{c_b^2} \text{cond}(\mathbf{A})$.

Both data-driven and residual-based optimization are potentially **ill-conditioned** when they are posed in a finite-dimensional trial space.

Condition Number and Optimization



Ill-conditioned optimization



Well-conditioned optimization

This talk:

- ▶ preconditioning for neural network training via [multilevel frames](#), and
- ▶ numerically stable implementation.

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Model problem

Consider the linear elliptic equation

$$-\operatorname{div}(a_y \nabla u(y)) = f, \quad \text{in } \Omega, \quad u = 0, \quad \text{on } \partial\Omega,$$

where a_y is bounded and strictly positive uniformly with respect to $y \in Y$, i.e.,

$$0 < \underline{a} \leq a_y(x) \leq \bar{a} < \infty, \quad \forall x \in \Omega, y \in Y.$$

Weak formulation

Find $u : Y \rightarrow U$ such that

$$\langle \mathcal{D}u(y), \mathcal{D}v \rangle_y = \langle \ell, v \rangle, \quad \forall v \in U, y \in Y.$$

- ▶ The parameter-independent map $\mathcal{D} : U \rightarrow \operatorname{range}(\mathcal{D}) \subset (L_2(\Omega))^m$ is an isomorphism.
- ▶ $\langle \cdot, \cdot \rangle_y$ is a bilinear form on $(L_2(\Omega))^m$ that may depend on y .

Example 1. For $U = H_0^1(\Omega)$, $\mathcal{D} = \nabla$, $m = d$, and $\langle \cdot, \cdot \rangle_y = \langle a_y \cdot, \cdot \rangle_{(L_2(\Omega))^m}$, the weak formulation reads: find $u : Y \rightarrow H_0^1(\Omega)$ such that

$$\langle a_y \nabla u, \nabla v \rangle_{(L_2(\Omega))^d} = \langle f, v \rangle, \quad \forall v \in H_0^1(\Omega), \quad y \in Y.$$

The solution is the unique minimizer of the energy functional

$$\min_{u \in H_0^1(\Omega)} \left\{ \frac{1}{2} \langle a_y \nabla u, \nabla u \rangle_{L_2(\Omega)} - \langle f, u \rangle \right\}. \quad (\text{E})$$

Variational Formulations

Example 2. Introducing an auxiliary variable $\sigma := a_y \nabla u$, the first-order system reads

$$a_y^{-1} \sigma - \nabla u = 0, \quad -\operatorname{div} \sigma = f, \quad \text{in } \Omega, \quad u = 0, \quad \text{on } \partial\Omega.$$

The weak formulation: for any $(v, \tau) \in H_0^1(\Omega) \times H(\operatorname{div}; \Omega)$,

$$\left\langle \begin{pmatrix} a_y^{-1} \sigma - \nabla u \\ -\operatorname{div} \sigma \end{pmatrix}, \begin{pmatrix} a_y^{-1} \tau - \nabla v \\ -\operatorname{div} \tau \end{pmatrix} \right\rangle_{(L^2(\Omega))^{2d+1}} = \left\langle \begin{pmatrix} 0 \\ f \end{pmatrix}, \begin{pmatrix} a_y^{-1} \tau - \nabla v \\ -\operatorname{div} \tau \end{pmatrix} \right\rangle_{(L^2(\Omega))^{2d+1}}.$$

Here $U = H_0^1(\Omega) \times H(\operatorname{div}; \Omega)$, $m = 2d + 1$, $\mathcal{D}(v, \tau) := (\nabla v, \tau, \operatorname{div} \tau)$, and

$$\langle (g, \tau, p), (h, \rho, q) \rangle_y := \langle p, q \rangle_{L_2} + \langle a_y^{-1} \tau - g, a_y^{-1} \rho - h \rangle_{(L^2(\Omega))^d}.$$

The solution minimizes the first-order system least squares (FOSLS) functional

$$\min_{(u, \sigma) \in H_0^1(\Omega) \times H(\operatorname{div}; \Omega)} \left\{ \|\operatorname{div} \sigma - f\|_{L^2(\Omega)}^2 + \|a_y^{-1} \sigma - \nabla u\|_{(L^2(\Omega))^d}^2 \right\}. \quad (\text{LS})$$

Hybrid Approximation and Stiffness Matrix

Hybrid approximation

$$u(x, y) \approx \sum_{i \in \mathcal{I}} u_i^{\text{NN}}(y; \theta) \varphi_i(x), \quad \forall (x, y) \in \Omega \times Y.$$

Training objective

$$\begin{aligned} \mathcal{L}(\theta) &= \frac{1}{K} \sum_{k=1}^K \left\{ \frac{1}{2} \langle \mathcal{D}u(y_k; \theta), \mathcal{D}u(y_k; \theta) \rangle_{y_k} - \langle \ell, u(y_k; \theta) \rangle_{L_2} \right\} \\ &= \frac{1}{K} \sum_{k=1}^K \left\{ \frac{1}{2} \langle \mathbf{A}(y_k) \mathbf{u}^{\text{NN}}(y_k; \theta), \mathbf{u}^{\text{NN}}(y_k; \theta) \rangle_{\ell_2} - \langle \mathbf{f}, \mathbf{u}^{\text{NN}}(y_k; \theta) \rangle_{\ell_2} \right\}. \end{aligned}$$

Stiffness matrix

$$\mathbf{A}(y) = (\langle \mathcal{D}\varphi_i, \mathcal{D}\varphi_{i'} \rangle_y)_{i, i' \in \mathcal{I}}.$$

Condition Number of the Stiffness Matrix

Stiffness matrix

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Standard finite element basis.

If Φ is the standard finite element basis with mesh size $h > 0$, then $\text{cond}(\mathbf{A}(y)) \lesssim h^{-2}$.

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If Φ is the standard finite element basis with mesh size $h > 0$, then $\text{cond}(\mathbf{A}(y)) \lesssim h^{-2}$.

Riesz basis.

If Φ is a Riesz basis, i.e., the frame operator

$$F : \ell_2(\mathcal{I}) \rightarrow U, \quad \mathbf{v} \mapsto \sum_{i \in \mathcal{I}} v_i \varphi_i,$$

is an isomorphism from $\ell_2(\mathcal{I})$ to $U_\Phi := \overline{\text{span}}(\Phi)$, then $\text{cond}(\mathbf{A}(y)) \lesssim 1$.

Examples: wavelet or Fourier bases $\rightsquigarrow$ difficult to construct in general domains.

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Frame.

Relaxing the Riesz basis condition by allowing Φ to be overcomplete. The frame operator F is an isomorphism from $\ker(F)^\perp$ to U_Φ .

- Hierarchy of finite element subspaces

$$W_1 \subset W_2 \subset \cdots \subset W_J.$$

- H^1 -normalized Lagrange basis for each level:

$$W_j := \text{span} \{ \varphi_{j,k} : k \in \mathcal{I}_j \}.$$

Theorem

Let $\|\varphi_{j,k}\|_{H^1(\Omega)} \approx 1$. Then

$$\{ \varphi_{j,k} : k \in \mathcal{I}_j, j = 1, \dots, J \}$$

is a frame for $W_J \subset H^1(\Omega)$ with frame constants independent of J .

Multilevel Frames: Preconditioned Loss

- Hybrid approximation with respect to the frame:

$$u(x, y) \approx (F\mathbf{w}^{\text{NN}}(y; \theta))(x) = \sum_{j=1}^J \sum_{k \in \mathcal{I}_j} w_{j,k}^{\text{NN}}(y; \theta) \varphi_{j,k}(x).$$

- Let $\mathbf{H}$ map frame coefficients to W_J -basis coefficients:

$$\mathbf{H} : \mathbf{w} \mapsto \mathbf{v}, \quad \text{where} \quad \sum_{j=1}^J \sum_{k \in \mathcal{I}_j} w_{j,k} \varphi_{j,k} = \sum_{k \in \mathcal{I}_J} v_k \varphi_{J,k}.$$

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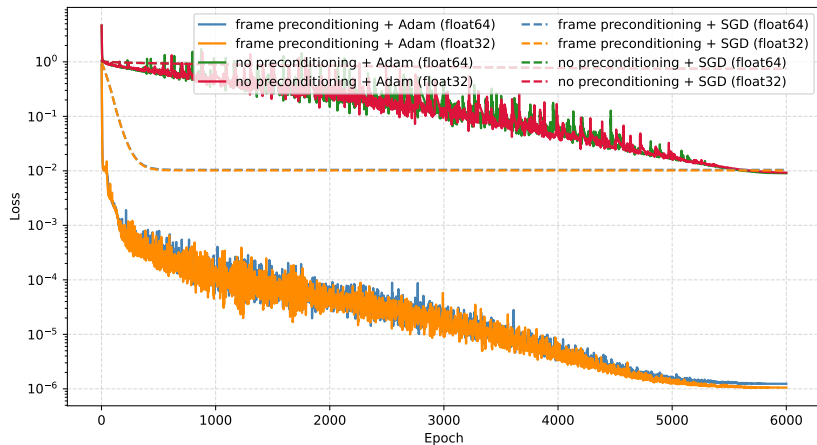
$$\mathbf{H} : \mathbf{w} \mapsto \mathbf{v}, \quad \text{where} \quad \sum_{j=1}^J \sum_{k \in \mathcal{I}_j} w_{j,k} \varphi_{j,k} = \sum_{k \in \mathcal{I}_J} v_k \varphi_{J,k}.$$

$$\mathcal{L}_{\text{frame}}(\theta) = \frac{1}{K} \sum_{k=1}^K \left\{ \frac{1}{2} \langle \mathbf{H}^\top \mathbf{A}(y_k) \mathbf{H} \mathbf{w}^{\text{NN}}(y_k; \theta), \mathbf{w}^{\text{NN}}(y_k; \theta) \rangle_{\ell_2} - \langle \mathbf{f}, \mathbf{H} \mathbf{w}^{\text{NN}}(y_k; \theta) \rangle_{\ell_2} \right\}.$$

Stiffness matrix with respect to the frame

$$\text{cond}(\mathbf{H}^\top \mathbf{A}(y) \mathbf{H}) = \text{cond}\left(\left(\langle \mathcal{D} \varphi_{j,k}, \mathcal{D} \varphi_{j',k'} \rangle_y\right)_{(j,k),(j',k')}\right) \lesssim 1.$$

Effect of Preconditioning



Effect of Preconditioning

Table: Comparison of FOSLS with and without preconditioning.

Optimizer	Precond.	Precision	RMSE of flux		RMSE of solution		MSE	Loss
			L_2 -norm	H^1 -norm	L_2 -norm	$H(\text{div})$ -norm		
SGD	✗	float32	2.70×10^{-1}	8.99×10^{-1}	9.91×10^{-2}	3.48×10^{-1}	9.30×10^{-1}	7.46×10^{-1}
	✗	float64	2.70×10^{-1}	8.99×10^{-1}	9.91×10^{-2}	3.48×10^{-1}	9.30×10^{-1}	7.47×10^{-1}
	✓	float32	4.64×10^{-2}	4.78×10^{-2}	2.01×10^{-2}	8.74×10^{-2}	9.93×10^{-3}	1.06×10^{-2}
	✓	float64	4.62×10^{-2}	4.76×10^{-2}	2.01×10^{-2}	8.74×10^{-2}	9.91×10^{-3}	1.06×10^{-2}
Adam	✗	float32	4.31×10^{-2}	4.92×10^{-2}	2.38×10^{-2}	8.54×10^{-2}	9.72×10^{-3}	9.12×10^{-3}
	✗	float64	4.28×10^{-2}	4.79×10^{-2}	2.29×10^{-2}	8.34×10^{-2}	9.25×10^{-3}	8.83×10^{-3}
	✓	float32	3.59×10^{-4}	4.90×10^{-4}	1.40×10^{-4}	9.32×10^{-4}	1.11×10^{-6}	1.19×10^{-6}
	✓	float64	4.42×10^{-4}	5.26×10^{-4}	1.37×10^{-4}	8.45×10^{-4}	9.90×10^{-7}	1.14×10^{-6}
L-BFGS	✗	float32	4.31×10^{-2}	4.48×10^{-2}	2.05×10^{-2}	8.87×10^{-2}	9.87×10^{-3}	1.04×10^{-2}
	✗	float64	4.31×10^{-2}	4.43×10^{-2}	2.02×10^{-2}	8.35×10^{-2}	8.93×10^{-3}	9.34×10^{-3}
	✓	float32	1.64×10^{-3}	2.51×10^{-3}	7.50×10^{-4}	5.29×10^{-3}	3.43×10^{-5}	3.43×10^{-5}
	✓	float64	5.10×10^{-4}	8.48×10^{-4}	1.77×10^{-4}	1.45×10^{-3}	2.82×10^{-6}	2.93×10^{-6}
NGD	✗	float32	2.81×10^{-1}	9.40×10^{-1}	9.91×10^{-2}	3.33×10^{-1}	9.95×10^{-1}	8.05×10^{-1}
	✗	float64	2.81×10^{-1}	9.40×10^{-1}	9.91×10^{-2}	3.33×10^{-1}	9.95×10^{-1}	8.06×10^{-1}
	✓	float32	1.47×10^{-3}	1.68×10^{-3}	3.95×10^{-4}	2.93×10^{-3}	1.14×10^{-5}	1.20×10^{-5}
	✓	float64	1.08×10^{-3}	1.28×10^{-3}	2.62×10^{-4}	2.02×10^{-3}	5.70×10^{-6}	6.08×10^{-6}

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Numerical Instability in Vanilla Multilevel Frame

Well-conditioned stiffness matrix.

$$\text{cond}(\mathbf{H}^\top \mathbf{A}(y) \mathbf{H}) \lesssim 1.$$

Numerical instability in implementation.

$$\mathbf{w} \xrightarrow[\text{cond}(\mathbf{H}) \gg 1]{\text{precision loss}} \mathbf{H}\mathbf{w} \xrightarrow[\text{cond}(\mathbf{A}(y)) \gg 1]{\text{precision loss}} \mathbf{A}(y)\mathbf{H}\mathbf{w} \xrightarrow[\text{cond}(\mathbf{H}) \gg 1]{\text{precision loss}} \mathbf{H}^\top \mathbf{A}(y) \mathbf{H}\mathbf{w}$$

- Half-precision (fp16) training fails due to rounding errors.

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- Half-precision (fp16) training fails due to rounding errors.

How can we construct a numerically stable implementation for preconditioning?

Stable Representations

- ▶ F is the frame operator
- ▶ $\mathcal{D}$ is the operator of vector-valued weak derivatives

$$\mathbf{w} \xrightarrow{F: \ell_2(\mathcal{I}) \rightarrow U_\Phi \subset U} F\mathbf{w} \xrightarrow{\mathcal{D}: U \rightarrow H = (L_2(\Omega))^m} \mathcal{D}F\mathbf{w}$$

Define

$$H_\Phi := \text{range}(\mathcal{D}F) \subset H = (L_2(\Omega))^m.$$

- ▶ If U_Φ is a finite-dimensional subspace of piecewise polynomials, then H_Φ is also a subspace of piecewise polynomials.

Proposition

- ▶ $\mathcal{D}F : \ell_2(\mathcal{I}) \rightarrow H_\Phi$ is uniformly well-conditioned as a mapping from $\ker(F)^\perp$ to H_Φ .
- ▶ The bilinear form $\langle \cdot, \cdot \rangle_y$ is bounded and elliptic on H_Φ uniformly in $y \in Y$.

Reformulation of the vanilla implementation.

- ▶ A naive basis $\{\mathcal{D}\varphi_{J,k} : k \in \mathcal{I}_J\}$ of H_Φ .
- ▶ Let $E : H_\Phi \rightarrow \ell_2(\mathcal{I}_J)$ satisfy

$$h = \sum_{k \in \mathcal{I}_J} (Eh)_k \mathcal{D}\varphi_{J,k}, \quad \forall h \in H_\Phi.$$

- ▶ Stiffness matrix with ill-conditioned decomposition

$$(EDF)^\top (\langle \mathcal{D}\varphi_{J,k}, \mathcal{D}\varphi_{J,k'} \rangle_y)_{k,k'} (EDF) = \mathbf{H}^\top \mathbf{A}(y) \mathbf{H}.$$

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Choosing different bases of H_Φ leads to different decompositions of the stiffness matrix.

Stable Representations

Stable representations.

- ▶ Let $\{\chi_n : n \in \mathcal{N}\}$ be a **frame** for H_Φ .
- ▶ Let $G : H_\Phi \rightarrow \ell_2(\mathcal{N})$ be an injective uniformly well-conditioned from H_Φ to $\text{range}(G)$ with

$$h = \sum_{n \in \mathcal{N}} (Gh)_n \chi_n, \quad \forall h \in H_\Phi.$$

- ▶ Stiffness matrix with well-conditioned decomposition

$$(GDF)^\top (\langle \chi_n, \chi_{n'} \rangle_y)_{n,n'} (GDF) = \mathbf{C}^\top \mathbf{D}(y) \mathbf{C}.$$

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- ▶ **Mathematical equivalence:**

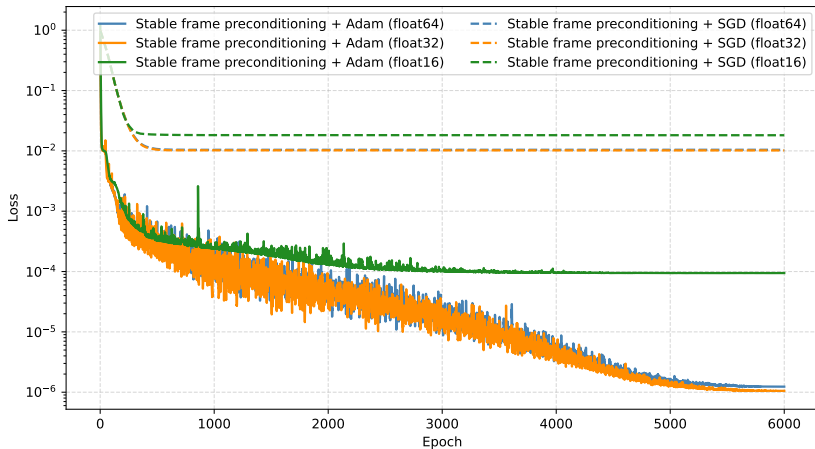
$$\mathbf{H}^\top \mathbf{A}(y) \mathbf{H} = \mathbf{D}^\top \mathbf{C}(y) \mathbf{D} \implies \text{cond}(\mathbf{D}^\top \mathbf{C}(y) \mathbf{D}) \lesssim 1.$$

- ▶ **Numerical stability:** Both $\mathbf{D}$ and $\mathbf{C}(y)$ are well-conditioned.

Outline

- ① Neural Network Methods for Parametric PDEs
- ② Frame Representation and Condition number
- ③ Numerically Stable Preconditioning
- ④ Numerical Experiments
- ⑤ Conclusions

Experimental Results



Experimental Results

Table: Comparison of FOSLS with stable preconditioning using different optimizers under different computing precisions.

Optimizer	Precision	RMSE of flux		RMSE of solution		MSE	Loss
		L_2 -norm	H^1 -norm	L_2 -norm	$H(\text{div})$ -norm		
SGD	float16	5.55×10^{-2}	1.11×10^{-1}	2.56×10^{-2}	1.02×10^{-1}	2.28×10^{-2}	1.86×10^{-2}
	float32	4.64×10^{-2}	4.78×10^{-2}	2.01×10^{-2}	8.74×10^{-2}	9.93×10^{-3}	1.06×10^{-2}
	float64	4.62×10^{-2}	4.76×10^{-2}	2.01×10^{-2}	8.74×10^{-2}	9.91×10^{-3}	1.06×10^{-2}
Adam	float16	2.39×10^{-3}	7.53×10^{-3}	9.36×10^{-4}	6.95×10^{-3}	1.05×10^{-4}	1.02×10^{-4}
	float32	3.57×10^{-4}	4.93×10^{-4}	1.43×10^{-4}	9.41×10^{-4}	1.20×10^{-6}	1.13×10^{-6}
	float64	4.42×10^{-4}	5.25×10^{-4}	1.37×10^{-4}	8.44×10^{-4}	1.14×10^{-6}	9.88×10^{-7}
L-BFGS	float16	4.70×10^{-2}	6.02×10^{-2}	2.06×10^{-2}	9.13×10^{-2}	1.20×10^{-2}	1.21×10^{-2}
	float32	1.68×10^{-3}	2.17×10^{-3}	9.37×10^{-4}	6.28×10^{-3}	4.41×10^{-5}	4.31×10^{-5}
	float64	3.35×10^{-4}	4.09×10^{-4}	1.23×10^{-4}	9.39×10^{-4}	1.05×10^{-6}	1.19×10^{-6}
NGD	float16	1.17×10^{-2}	3.10×10^{-2}	4.07×10^{-3}	2.82×10^{-2}	1.76×10^{-3}	1.71×10^{-3}
	float32	1.15×10^{-3}	1.34×10^{-3}	3.63×10^{-4}	2.70×10^{-3}	9.09×10^{-6}	9.39×10^{-6}
	float64	9.73×10^{-4}	1.01×10^{-3}	2.73×10^{-4}	1.79×10^{-3}	4.22×10^{-6}	4.43×10^{-6}

Summary:

- ▶ Neural network training for parametric PDEs with a hybrid approximation leads to a quadratic optimization problem whose stiffness matrix can be severely ill-conditioned.
- ▶ Multilevel frames provide a theoretically clean preconditioner. However, the naive implementation of this preconditioner is numerically unstable.
- ▶ We propose a numerical stable implementation for preconditioning, enabling single- and half-precision training.

Reference

- [1] Markus Bachmayr, Wolfgang Dahmen, Chenguang Duan, and Mathias Oster. Preconditioning and Numerical Stability in Neural Network Training for Parametric PDEs. [arXiv:2601.06514](https://arxiv.org/abs/2601.06514)